

► CARDIOVASCULAR—EMBRYOLOGY

Heart morphogenesis First functional organ in vertebrate embryos; beats spontaneously by week 4 of development.

Cardiac looping Primary heart tube loops to establish left-right polarity; begins in week 4 of development.

Defect in left-right **d**ynein (involved in left-right asymmetry) can lead to **d**extrocardia, as seen in Kartagener syndrome.

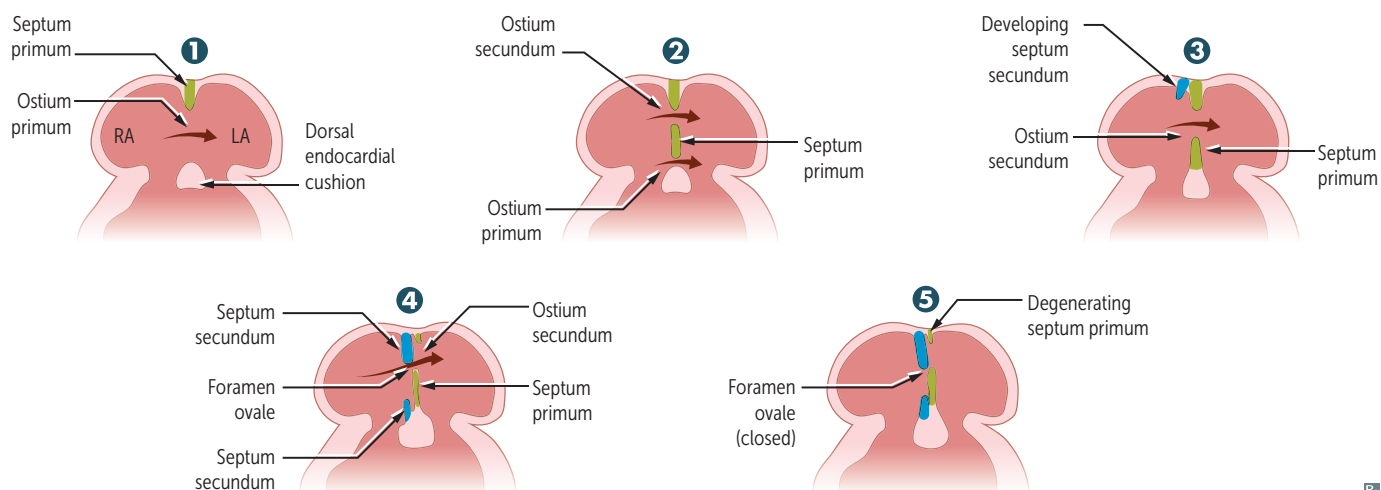
Septation of the chambers

Atria

- ❶ Septum primum grows toward endocardial cushions, narrowing ostium primum.
- ❷ Ostium secundum forms in septum primum due to cell death (ostium primum regresses).
- ❸ Septum secundum develops on the right side of septum primum, as ostium secundum maintains right-to-left shunt.
- ❹ Septum secundum expands and covers most of ostium secundum. The residual foramen is the foramen ovale.
- ❺ Remaining portion of septum primum forms the one-way valve of the foramen ovale.

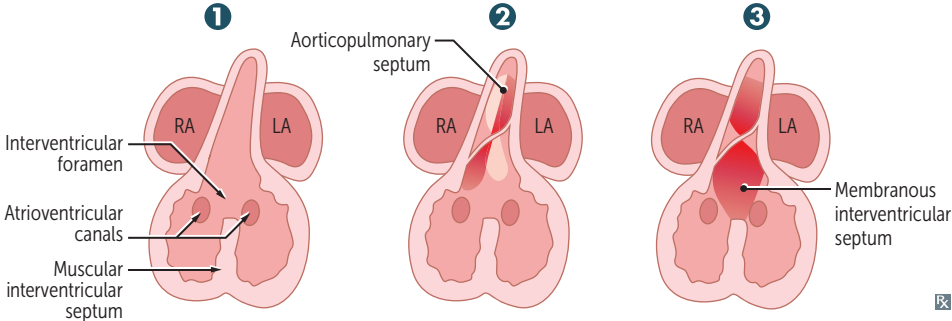
6. Septum primum closes against septum secundum, sealing the foramen ovale soon after birth because of ↑ LA pressure and ↓ RA pressure.
7. Septum secundum and septum primum fuse during infancy/early childhood, forming the atrial septum.

Patent foramen ovale—failure of septum primum and septum secundum to fuse after birth; seen in 25% of population. Most are asymptomatic and remain undetected. Can lead to paradoxical emboli (venous thromboemboli entering the systemic arterial circulation through right-to-left shunt) as can occur in atrial septal defect (ASD).



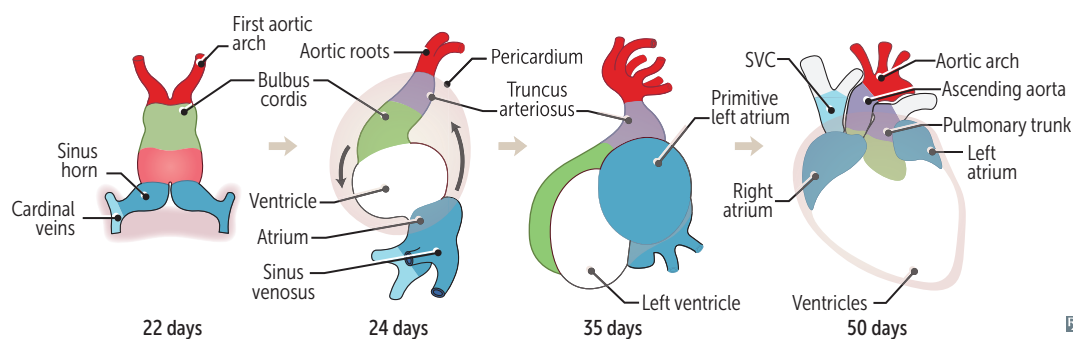
Heart morphogenesis (continued)

Ventricles	❶ Muscular interventricular septum forms. Opening is called interventricular foramen. ❷ Aorticopulmonary septum rotates and fuses with muscular ventricular septum to form membranous interventricular septum, closing interventricular foramen. ❸ Growth of endocardial cushions separates atria from ventricles and contributes to both atrial septation and membranous portion of the interventricular septum.	Ventricular septal defect—most common congenital cardiac anomaly, usually occurs in membranous septum. Atrioventricular septal defect (AVSD)—also known as endocardial cushion or AV canal defect. Acyanotic congenital heart defect with single common AV valve plus either ASD alone (partial AVSD) or both ASD and VSD (complete AVSD). Associated with Down syndrome, maternal diabetes, and obesity.
Outflow tract formation	Neural crest cell migrations → truncal and bulbar ridges that spiral and fuse to form aorticopulmonary septum → ascending aorta and pulmonary trunk.	Conotruncal abnormalities associated with failure of neural crest cells to migrate: ▪ Transposition of great arteries (TGA). ▪ Tetralogy of Fallot. ▪ Persistent truncus arteriosus.
Valve development	Aortic/pulmonary: derived from endocardial cushions of outflow tract. Mitral/tricuspid: derived from fused endocardial cushions of the AV canal.	Valvular anomalies may be stenotic, regurgitant, atretic (eg, tricuspid atresia), or displaced (eg, Ebstein anomaly).



Heart embryology

EMBRYONIC STRUCTURE	GIVES RISE TO
Right common cardinal vein and right anterior cardinal vein	Superior vena cava (SVC)
Posterior cardinal, subcardinal, and supracardinal veins	Inferior vena cava (IVC)
Right horn of sinus venosus	Smooth part of right atrium (sinus venarum)
Left horn of sinus venosus	Coronary sinus
Primitive pulmonary vein	Smooth part of left atrium
Primitive atrium	Trabeculated part of left and right atria
Endocardial cushion	Atrial septum, membranous interventricular septum; AV and semilunar valves
Primitive ventricle	Trabeculated part of left and right ventricles
Bulbus cordis	Smooth parts (outflow tract) of left and right ventricles
Truncus arteriosus	Ascending aorta and pulmonary trunk



Fetal-postnatal derivatives

FETAL STRUCTURE	POSTNATAL DERIVATIVE	NOTES
Ductus arteriosus	Ligamentum arteriosum	Near the left recurrent laryngeal nerve
Ductus venosus	Ligamentum venosum	
Foramen ovale	Fossa ovalis	
Allantois → urachus	Median umbilical ligament	Urachus is part of allantois between bladder and umbilicus
Umbilical arteries	Medial umbilical ligaments	
Umbilical vein	Ligamentum teres hepatis (round ligament)	Contained in falciform ligament

Fetal circulation

3 important shunts:

1. Umbilical vein → **ductus venosus** → IVC (bypassing hepatic circulation).
2. Most of the highly **oxygenated** blood from IVC → **foramen ovale** → LA.
3. Deoxygenated blood from SVC → RA → RV → main pulmonary artery → **ductus arteriosus** → **descending aorta**; shunt is due to ↑ fetal pulmonary artery resistance.

At birth, infant takes a breath → ↓ resistance in pulmonary vasculature → ↑ LA pressure vs RA pressure → foramen ovale closes (now called fossa ovalis); ↑ in O₂ (from respiration) and ↓ in prostaglandins (from placental separation) → closure of ductus arteriosus.

NSAIDs (eg, indomethacin, ibuprofen) or acetaminophen help close the patent ductus arteriosus (PDA) → ligamentum arteriosum (remnant of ductus arteriosus). “**End**methacin” **ends** the PDA. Prostaglandins **E**₁ and **E**₂ **kEEp** PDA open.

